

carbon capture, utilization and sequestration (CCUS)

Module 5 Lecture 3

Carbon Capture, Utilization and Sequestration (CCUS)

- **CO₂ is captured before entering the atmosphere and then either permanently stored underground or incorporated into certain types of products, such as concrete or chemicals.**
- **The main sectors where CCUS can be deployed are power and industry**
- **CCUS is not the same as carbon removal, which removes CO₂ from the atmosphere**

The International Energy Agency (IEA) defines Carbon Capture, Utilization and Storage (CCUS) as a group of technologies for capturing of CO₂ from large and stationary CO₂ emitting sources, such as fossil fuel-based power plants and other industries. CCUS also involves the transport of the captured CO₂ (typically by pipeline and in certain situations by through shipping, rail or trucks also) to sites, either for utilization in different applications or injection into geological formations or depleted oil & gas fields for permanent storage and trapping of the CO₂.

Carbon Removal

- Natural carbon removal through reforestation and forest management
- Direct air capture is a technology that uses chemical reactions to pull carbon dioxide out of air using either liquid solvents or solid sorbents.
- Biomass carbon removal and storage (BiCRS) includes a range of processes that use biomass from plants or algae to remove carbon dioxide from the air and then store it for long periods of time.
- Some minerals naturally react with CO₂, turning carbon dioxide from a gas into a solid and keeping it out of the atmosphere permanently. This process is commonly referred to as “carbon mineralization” or “enhanced weathering,”
- Ocean-based carbon removal approaches have been proposed to leverage the ocean’s capacity to sequester

Differentiating carbon capture and carbon removal

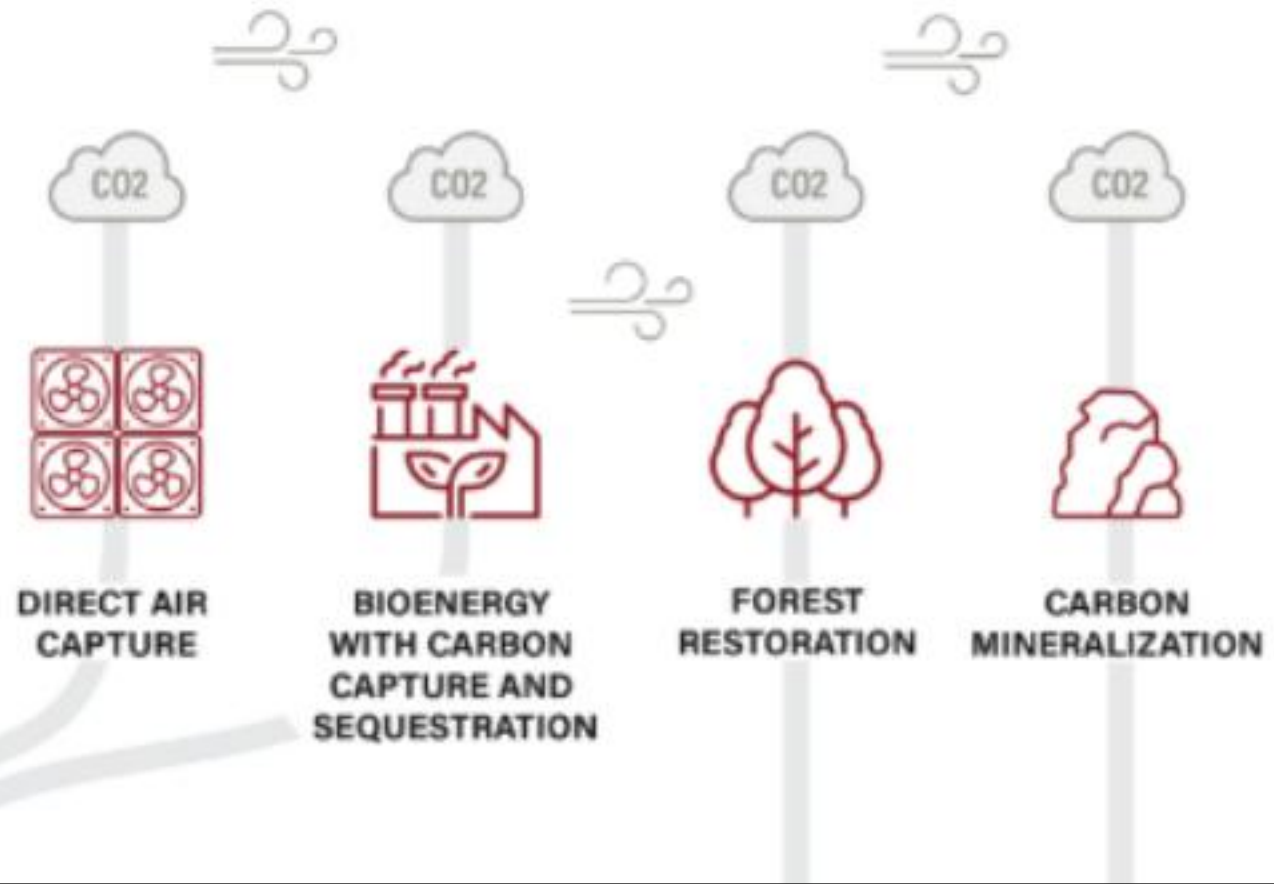
CARBON CAPTURE, UTILIZATION AND SEQUESTRATION (CCUS)

Capture of CO₂ at emissions source
(e.g., industrial or power facilities)
to prevent it from going into the air



CARBON DIOXIDE REMOVAL (CDR)

Activities that **remove** CO₂ that's already in the air, including the following



CCUS

- Captured CO₂ can be pumped underground into certain geological formations where it is permanently sequestered,
- CO₂ can be used in products ranging from concrete to chemicals to synthetic fuels.
- If used in products, the duration of sequestration depends on the product
- For example, if CO₂ is used to produce synthetic fuel, it would be re-emitted when the fuel is combusted, while CO₂ used in concrete would be sequestered permanently.

CCUS

- There are around 40 operational CCUS projects globally, with about 25 under construction
- Operational projects are capturing between 42 and 49 million metric tons of CO₂ per year
- (MtCO₂/yr).

CCUS projects by stage of development and capture capacity

Global CCS Institute International Energy Agency

Number of projects



Capture capacity (MtCO₂/yr)



Source: Global CCS Institute (GCCSI) and IEA. • Note: GCCSI data as of July 2023; IEA numbers as of March 2023. IEA numbers exclude DAC projects and projects that just include transport and/or sequestration of CO₂.



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CCUS Technologies

- Chemical solvent-based CO₂ capture technologies
 - Preferred when dealing with gas streams that are lean in CO₂ and have relatively lower pressures, such as flue gas streams from power plants, blast furnace gases in steel plants, gas streams in refineries or chemicals plants.
 - The cost and availability of steam is also a key factor as regenerating the solvent requires large quantities of steam.
- Physical solvent-based CO₂ capture technologies
 - These work well on gas streams with relatively higher CO₂ concentration and pressure, such as pre-combustion capture in the case of gasification projects.

CCUS Technologies

- Adsorption-based CO₂ capture
 - Suitable for gas streams with moderate to high pressure and moderate CO₂ concentration such as Steam methane reforming flue gas or blast furnace gas.
- Cryogenic CO₂ capture
 - Preferred in cases where the cost of power is low. This technology can be applied for carbon capture from the Pressure Swing Adsorption tail gas of Steam Methane Reforming Units (for producing H₂) and provides a unique advantage of increasing the yield/recovery of hydrogen production from the same quantity of feedstock (natural gas).

Converting Captured CO₂ to Value Added Products

- **Green urea:** Green urea can be produced from the captured CO₂ and cost-competitive green hydrogen, from renewable energy-based electrolysis of water.
- **F&B applications:** CO₂ is utilized in F&B applications such as carbonated drinks, dry ice, and modified atmosphere packing
- **Building materials (concrete and aggregates):** Utilizing CO₂ for producing building materials through concrete curing and aggregate formation. In these applications, CO₂ is injected in a liquid state without any conversion, thus reducing the energy requirements.
- **Chemicals (methanol and ethanol):** Conversion of CO₂ to methanol and ethanol
- **Polymers (including bio-plastics)**
- **Enhanced Oil Recovery (EOR):** CO₂ based EOR has been successfully operating for decades for producing low-carbon oil from maturing oil fields

Limitations of CCUS system

- High upfront costs (often upwards of \$1 billion)
- Today's carbon capture systems do not capture 100% of emissions. Most are designed to capture 90%, but reported capture rates are lower in some case
- Transport and geologic sequestration of CO₂ also present risks — mainly of CO₂ leakage.
- Unlike many other clean technologies (such as solar photovoltaic), CCUS systems can't be mass produced because they are specifically designed to match the facility that's capturing the CO₂
- CCUS projects are also complex to coordinate because each step of the process — capture, transport and sequestration — is often owned and operated by a different company